CSCC63 --- Computability & computational complexity

Week 5 Tutorial

Practice with reductions

Question 1

Consider the language

$$A = \{\langle M \rangle : |\mathcal{L}(M)| \geq 3\}$$

(i.e., the set of codes of TMs that accept at least three inputs).

Is A decidable?
If not, is it recognizable?
What about $\bar{A}$?
Justify your answers.

Question 1 (cont'd)

$$A = \{\langle M\rangle : |\mathcal{L}(M)| \geq 3\}$$

Is A decidable?

A is undecidable.

Follows immediately from Rice's Theorem, since the property of a recognizable language having at least 3 elements is nontrivial (some do and some don't).

Question 1 (cont'd)

$$A = \{\langle M \rangle : |\mathcal{L}(M)| \geq 3\}$$

Is A decidable?

Prove undecidability without using Rice's Theorem.
Prove that $UNIV \leq_m A$.

Want a computable function f: $\langle M, x \rangle \longmapsto \langle M' \rangle$ s.t.

M accepts x $\implies$ M' accepts at least 3 inputs
M doesn't accept x $\implies$ M' accepts fewer than 3 inputs

Question 1 (cont'd)

$$A = \{\langle M \rangle : |\mathcal{L}(M)| \geq 3\}$$

all

Want computable f: $\langle M, x \rangle \longmapsto \langle M' \rangle$ s.t.

M accepts x $\Longrightarrow$ M' accepts at least 3 inputs

M doesn't accept x $\Longrightarrow$ M' accepts fewer than 3 inputs

no

$M' :=$ **on input** y:

  run M on x [uses universal TM]

  if M accepts x **then** accept y

  else reject y

Transformation of $\langle M, x \rangle$ to $\langle M' \rangle$ clearly computable.

Question 1 (cont'd)

$$A = \{\langle M \rangle : |\mathcal{L}(M)| \geq 3\}$$

Since A is not decidable, next question:
Is A recognizable?

Yes.
Argument 1: When considering an input $\langle M \rangle$, dovetail through pairs (i, j) running M on the i-th input x_i (in shortlex order) for j steps. If we find such pairs for which M accepts three distinct inputs, we accept $\langle M \rangle$.

Question 1 (cont'd)

$$A = \{\langle M \rangle : |\mathcal{L}(M)| \geq 3\}$$

A recognizer for A in pseudocode:

on input $\langle M \rangle$:
$S := \emptyset$
for each pair (i, j) in dovetail order **do**
  run M on x_i for j steps
  if M accepts x_i in j steps then $S := S \cup \{x_i\}$
  if $|S| = 3$ then accept $\langle M \rangle$

On your own: Why does this recognize L? Why does it not decide L?

Question 1 (cont'd)

$$A = \{\langle M \rangle : |\mathcal{L}(M)| \geq 3\}$$

Is A recognizable?

Yes.

Argument 2: A is recognized by a nondeterministic TM that on input $\langle M \rangle$ guesses three different inputs for M and verifies that M accepts them. If so, the NTM accepts $\langle M \rangle$.

How does a NTM "guess" a string?

Question 1 (cont'd)

$$A = \{\langle M \rangle : |\mathcal{L}(M)| \geq 3\}$$

Pseudocode for NTM that recognizes A

on input $\langle M \rangle$:
guess three strings x_1, x_2, x_3 bit by bit
if $x_1 \neq x_2$ **and** $x_1 \neq x_3$ **and** $x_2 \neq x_3$ **then**
  run M on x_1
  if M accepts x_1 **then**
    run M on x_2
    if M accepts x_2 **then**
      run M on x_3
      if M accepts x_3 **then** accept $\langle M \rangle$

Question 1 (cont'd)

$$A = \{\langle M \rangle : |\mathcal{L}(M)| \geq 3\}$$

Why does this NTM accept $\langle M \rangle$ if and only if $\langle M \rangle \in A$?

[$\Longleftarrow$]: If $\langle M \rangle \in A$, then some computation branch of the NTM will guess distinct x_1, x_2, x_3 in $\mathcal{L}(M)$ and will verify that M accepts them, so the NTM will accept $\langle M \rangle$.

[$\Longrightarrow$]: If $\langle M \rangle \notin A$, then no computation branch of the NTM can guess distinct x_1, x_2, x_3 and verify that M accepts them. All computation branches will reject or loop (as they keep guessing bits forever), so the NTM will not accept $\langle M \rangle$.

Question 1 (cont'd)

$$A = \{\langle M \rangle : |\mathcal{L}(M)| \geq 3\}$$

We have seen that A is undecidable but recognizable.
Next question:
What about $\bar{A}$? Is it decidable? If not, is it recognizable?

$\bar{A}$ is not decidable: If it were, so would its complement A (Theorem 3.3).

$\bar{A}$ is not recognizable: If it were, since its complement is also recognizable, both would be decidable (Theorem 3.6).

Question 2

Consider the language

$$B = \{\langle M\rangle : |\mathcal{L}(M)| = 3\}$$

(i.e., the set of codes of TMs that accept exactly three inputs).

Is B decidable?

If not, is it recognizable?

What about $\overline{B}$?

Justify your answers.

Question 2 (cont’d)

$$B = \{\langle M \rangle : |\mathcal{L}(M)| = 3\}$$

Consider B and $\bar{B}$.
Are they decidable?

No, by Rice’s Theorem: A recognizable language having exactly three elements is a nontrivial property (some do and some don’t).

Question 2 (cont'd)

$$B = \{\langle M\rangle : |\mathcal{L}(M)| = 3\}$$

Is either B or $\bar{B}$ recognizable?
Try the standard approaches.

They don't seem to work --- e.g., consider dovetailing:

- Can't stop if we have found fewer than 3 inputs that M accepts and thereby reject $\langle M\rangle$: there might be more and in fact $\langle M\rangle \in B$.
- Can't stop if we have found 3 and thereby accept $\langle M\rangle$: there might be more and in fact $\langle M\rangle \notin B$.
- Can stop if we find more then 3 inputs and reject $\langle M\rangle$. But these are not the only "no" instances.

Question 2 (cont'd)

$$B = \{\langle M \rangle : |\mathcal{L}(M)| = 3\}$$

So maybe B and $\bar{B}$ are not recognizable.
Try proving:

- $UNIV \leq_m B$. This shows that $\bar{B}$ is not recognizable.
- $UNIV \leq_m \bar{B}$. This shows that B is not recognizable.

Do these on your own outside of tutorial.

Question 3

Consider the language "writes symbol on empty tape"

$WS = \{\langle M, a\rangle : \text{TM } M \text{ on empty tape writes symbol } a\}$

Is WS decidable? If not, is it recognizable?

Question 3 (cont'd)

$$WS = \{\langle M, a\rangle : \text{TM } M \text{ on empty tape writes symbol } a\}$$

Is WS decidable?

We cannot use Rice's Theorem: This is not about the language accepted by TMs, but about the behaviour of TMs.

Question 3 (cont'd)

$WS = \{\langle M, a\rangle : \text{TM } M \text{ on empty tape writes symbol } a\}$

Is WS decidable?

This language is undecidable.

Recall the language

$$HET = \{\langle M\rangle : \text{TM } M \text{ halts on empty tape}\}$$

Prove that $HET \leq_m WS$.
Since HET is undecidable, so is WS.

Question 3 (cont'd)

$$WS = \{\langle M, a\rangle : \text{TM } M \text{ on empty tape writes symbol } a\}$$

Given $\langle M\rangle$ produce $\langle M', a\rangle$ s.t.
M halts on empty tape $\Longrightarrow$ M' writes a on empty tape
M doesn't halt on empty tape $\Longrightarrow$ M' doesn't write a on empty tape

M' is like M except that

- it has a new tape symbol, say @
- when M has a transition to the accept or reject state, M' has a similar transition except that it writes @

Question 3 (cont'd)

$WS = \{\langle M, a\rangle : \text{TM } M \text{ on empty tape writes symbol } a\}$

More formally: If the transition function of M contains $\delta(q, a) = (p, b, D)$ for any state q, symbols a, b, direction D, and state $p \in \{h_A, h_R\}$, then the transition function δ' of M' has instead $\delta'(q, a) = (p, @, D)$. All other transitions in M' are exactly as in M.

Transformation $\langle M\rangle \longmapsto \langle M', @\rangle$ clearly computable

This ensures:
M halts on empty tape $\Longleftrightarrow$ M' writes @ on empty tape

So $HET \leq_m WS$.

Question 3 (cont'd)

$WS = \{\langle M, a\rangle : \text{TM } M \text{ on empty tape writes symbol } a\}$

Since WS is not decidable, on to the next question:
Is it recognizable?

Do this on your own, after the tutorial.

Question 4

Consider the language "writes non-blank symbol on empty tape"

$$WNB = \{\langle M \rangle : \text{TM } M \text{ on empty tape writes some non−blank symbol}\}$$

Is WNB decidable? If not, is it recognizable?

Question 4 (cont'd)

$$WNB = \{\langle M \rangle : \text{TM } M \text{ on empty tape writes some non–blank symbol}\}$$

Is WNB decidable?

Yes, it is:

<u>WNB-Decider($\langle M \rangle$)</u>:
$k :=$ # of states of M
run M on empty tape for $k+1$ steps (or until it halts)
if M writes $a \neq \sqcup$ during these steps **then** accept $\langle M \rangle$
else reject $\langle M \rangle$

Question 4 (cont'd)

WNB-Decider($\langle M \rangle$):
$k := \#$ of states of M
run M on empty tape for $k+1$ steps (or until it halts)
if M writes $a \neq \sqcup$ during these steps **then** accept $\langle M \rangle$
else reject $\langle M \rangle$

Why does this work?

If after $k+1$ steps M has not written a non-blank symbol, then during this computation it has repeated a state (pigeonhole principle!) and the tape is still blank. So we have a loop that repeats forever, without M ever writing a non-blank.

Question 4 (cont'd)

Proof that WNB is undecidable (!!!)

Given $\langle M \rangle$, produce a TM M' as follows:

M' is identical to M, except that it has a new tape symbol @, and whenever M takes a transition in which it writes a non-blank symbol, M' instead writes @ and accepts (or rejects --- it doesn't matter which).

So, M on empty tape writes a non-blank symbol
$\Leftrightarrow$ M' writes @ on empty tape.
Since we can't decide if a TM on empty tape writes a particular symbol (previous question), we can't decide if a TM on empty tape writes a non-blank symbol.

Question 4 (cont'd)

So...

- is the decider wrong? or
- is this proof wrong?

This proof is wrong.
The very last step is wrong.
We proved a correct reduction in the wrong direction that does not allow us to make the inference we want.
We proved

$$WNB \leq_m WS.$$

We know that WS is undecidable, but this does not mean that WNB is undecidable: it is a much easier problem!